

Here again, Eq. (9.2.1) exactly fits the form of the eigenvalue problem that was given by Eq. (8.1.28), and the **generalized eigenproblem for lateral motion** is

$$\begin{pmatrix} R_{y,\beta} & R_{y,\bar{p}} & (R_{y,\bar{r}} - 1) & 0 & R_{gy} \cos \theta_o & 0 \\ R_{\ell,\beta} & R_{\ell,\bar{p}} & R_{\ell,\bar{r}} & 0 & 0 & 0 \\ R_{n,\beta} & R_{n,\bar{p}} & R_{n,\bar{r}} & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & \cos \theta_o \\ 0 & 1 & \tan \theta_o & 0 & 0 & 0 \\ 0 & 0 & \sec \theta_o & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \chi_\beta \\ \chi_{\bar{p}} \\ \chi_{\bar{r}} \\ \chi_{\xi_y} \\ \chi_\phi \\ \chi_\psi \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & -t_{xz} & 0 & 0 & 0 \\ 0 & -t_{zx} & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \chi_\beta \\ \chi_{\bar{p}} \\ \chi_{\bar{r}} \\ \chi_{\xi_y} \\ \chi_\phi \\ \chi_\psi \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad (9.2.2)$$

The eigenvalues and eigenvectors associated with linearized lateral aircraft motion can be found by solving Eq. (9.2.2) using a generalized eigensolver. If a generalized eigensolver is not available, the system can easily be rearranged to give the **special eigenproblem for lateral motion**:

$$\begin{pmatrix} R_{y,\beta} & R_{y,\bar{p}} & (R_{y,\bar{r}} - 1) & 0 & R_{gy} \cos \theta_o & 0 \\ \frac{R_{\ell,\beta} + t_{xz} R_{n,\beta}}{1 - t_{xz} t_{zx}} & \frac{R_{\ell,\bar{p}} + t_{xz} R_{n,\bar{p}}}{1 - t_{xz} t_{zx}} & \frac{R_{\ell,\bar{r}} + t_{xz} R_{n,\bar{r}}}{1 - t_{xz} t_{zx}} & 0 & 0 & 0 \\ \frac{R_{n,\beta} + t_{zx} R_{\ell,\beta}}{1 - t_{xz} t_{zx}} & \frac{R_{n,\bar{p}} + t_{zx} R_{\ell,\bar{p}}}{1 - t_{xz} t_{zx}} & \frac{R_{n,\bar{r}} + t_{zx} R_{\ell,\bar{r}}}{1 - t_{xz} t_{zx}} & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & \cos \theta_o \\ 0 & 1 & \tan \theta_o & 0 & 0 & 0 \\ 0 & 0 & \sec \theta_o & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \chi_\beta \\ \chi_{\bar{p}} \\ \chi_{\bar{r}} \\ \chi_{\xi_y} \\ \chi_\phi \\ \chi_\psi \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \chi_\beta \\ \chi_{\bar{p}} \\ \chi_{\bar{r}} \\ \chi_{\xi_y} \\ \chi_\phi \\ \chi_\psi \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} \quad (9.2.3)$$